THE PATH:
THE MANUAL

A parallel engineering guide for ∞ and Starman Industries

Thirteen chapters and two expansions - every stage of the Starman Line, walked beside the companies that already ran it

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Chapter 0 - THE ORIGIN

> *Before the family builds its path, the chapter is about the two paths that came first. Microsoft and Apple both started as two people, one product, and a customer who almost did not exist. This chapter walks how each got started, with the arithmetic they ran and the metrics they made, because the parallel is the manual.*

Microsoft: the license, not the box

January 1975. Paul Allen walks through Harvard Square, picks up a newsstand copy of Popular Electronics, and finds the Altair 8800 on the cover - a build-it-yourself computer kit from a little Albuquerque company called MITS, with no screen, no keyboard, just lights and switches (OPB, Apr 2025). Allen runs to Bill Gates: we have to catch this train. The two had already tried a business together - Traf-O-Data, in 1972, a box that processed traffic-counter tapes. It barely sold. It taught them everything they used next (Wikipedia: Microsoft).

The move that made the company: they had no Altair, had never touched one. Gates wrote the BASIC interpreter against a simulator Allen built; Allen flew to Albuquerque in March 1975 and demonstrated it on a real Altair for the first time in his life. It worked flawlessly on the first run (Oreate AI Guides). MITS agreed to distribute it as Altair BASIC, and Micro-Soft was founded April 4, 1975, in Albuquerque, because that is where the customer was (Preceden timeline).

February 1976: Gates writes the Open Letter to Hobbyists, furious that people are copying the paper tape and not paying. It reads harsh and it is the founding document of paid software (CorpDigest: founders). By end of 1978 sales top $1 million; the company moves to Bellevue, Washington in 1979 (History.com).

The decisive deal, 1980: IBM needs an operating system for its new Personal Computer. Microsoft does not have one. Gates buys 86-DOS (QDOS) from Seattle Computer Products for $50,000, adapts it into MS-DOS, and licenses it to IBM - and here is the whole lesson - *keeps the right to license it to everyone else* (CorpDigest: history). When the clone market explodes, every clone ships Microsoft software. The 1986 IPO at $21 a share raises $61 million and eventually makes something like 12,000 employees millionaires; in 1987 Gates becomes the world's youngest billionaire (Wikipedia; Preceden).

Apple: the machine, then the killer app

April 1, 1976: Apple Computer Company, a partnership of Steve Jobs, Steve Wozniak, and Ronald Wayne. Wozniak has designed the Apple I around MOS Technology's $20 6502 chip - he had spent years designing computers on paper waiting for a CPU he could afford (Wikipedia: History of Apple). To finance the first run, Jobs sells his Volkswagen bus ($1,500) and Wozniak sells his HP-65 calculator (Mac History). The Apple I sells at $666.66 as a bare board. July 1976, the Homebrew Computer Club demo lands the Byte Shop's order for 50 fully assembled machines - the first real customer, secured before the company could build 50 of anything (Grokipedia: Apple).

Ronald Wayne holds 10 percent for twelve days, then sells his share back for $800 (about $4,530 in 2025 dollars), afraid of the liability (Wikipedia). Banks will not lend - a computer for ordinary people sounds absurd - so in August 1976 Nolan Bushnell points Jobs to Don Valentine, who passes, but hands him Mike Markkula. Markkula invests $250,000 at incorporation (January 3, 1977), recruits Michael Scott as CEO, and secures a Bank of America credit line (Wikipedia: Apple Inc.; EBSCO).

The Apple II ships June 1977 at $1,298 with color graphics. Then 1979: VisiCalc, the first spreadsheet, ships exclusively for the Apple II and becomes the first killer app - businesses buy 90 percent of Apple II units from then on (Wikipedia). The revenue curve: $774,000 in fiscal 1977 to $118 million in fiscal 1980, 152.5x in three years, an implied compound rate of about 434 percent a year; The Verge counts it as 640 percent growth in 1979 and 230 percent in 1980 (Fast Company; The Verge). The December 12, 1980 IPO raises more capital than any offering since Ford in 1956, values the company at $1.8 billion, and mints about 300 millionaires overnight (History of Apple).

The arithmetic the origin runs on

Fact	Number	Read
Apple revenue 1977 to 1980	$774,000 to $118,000,000	152.5x in three fiscal years; implied CAGR 434 percent
Doubling cadence, first five years	about every 4 months	9 doublings would take $774k to $396M - the real curve lands at 152x, so the doubling story holds roughly, then bends
Markkula's $250,000 at IPO	$203 million in stock	812x in under four years - the first check is the cheapest money any company ever spends
Wayne's 10 percent	sold for $800; worth $180,000,000 at IPO	225,000x - fear is the most expensive thing ever sold
QDOS purchase price	$50,000	the asset that licensed an industry; Microsoft's whole empire is one acquired file system plus one non-exclusive clause
Apple II units 1979 / 1980	35,000 / 78,000	revenue per unit in 1980 about $1,513 against a $1,298 base price - peripherals, disk drives, and the app tax were already in the mix

Microsoft's first sale to IPO gap	1975 to 1986, $1M to $61M raised	eleven years between the first customer and liquidity

What the parallel says to the family

Microsoft sold the license; Apple sold the machine; both were two people and one product within eighteen months of the magazine cover. Neither raised venture money first - Microsoft bootstrapped on Altair royalties, Apple bootstrapped on a bus, a calculator, and one 50-unit order. The check that mattered (Markkula, $250k) arrived only after a working product and a paying customer existed. ∞ copies the sequence, not the scale: product, customer, then capital. Chapter by chapter this manual walks the same shape - every stage of the Starman Line set beside the company that already ran it, with the numbers that company proved.

Chapter 1 - THE SUIT (parallel: Whoop, Oura, and the wearable ledger)

> *The Luna Suit chapter of the first book designed a washable, sensing, deep-pressure suit. The parallel industry already proved the business: the body-worn sensor, sold as a subscription.*

The precedent

Whoop, founded 2012 out of Harvard, sells no hardware: an annual membership includes the band, the app, and the coaching. By March 2026 it had 2.5 million members, a $1.1 billion annualized bookings run-rate at the end of 2025 (up 103 percent year over year), operating-cash-flow positivity, and a $575 million Series G at a $10.1 billion valuation (Sacra: Whoop; Circuly case study). Oura, founded 2013 in Oulu, Finland, sold about 5.5 million rings with roughly 5 million paying members on the way past $1 billion of 2025 revenue, about 80 percent hardware and 20 percent subscription, and filed its S-1 in September 2026 (Ringing the Bell; Sacra: Oura).

The arithmetic

Fact	Number	Read
Whoop bookings per member	$1.1B / 2.5M members	about $440 per member per year - subscription hardware is a payment plan for trust
Oura revenue per ring sold	$1B / 5.5M rings	about $182 lifetime so far, $500 per active subscriber
Wearable market	$56.8B smart jewelry in 2025, 7.5 percent CAGR	the tide the Luna Suit floats on (GM Insights)
Suit-side power (carried)	200 Wh Luna pack; EPIONE session 84 Wh	2.4 full heat-plus-TENS sessions per charge, script-verified

The path for the family

Starman Industries does not sell the Luna Suit; it warrants the pilot. Hardware at cost, comfort and data as the membership, ThreadCat holding the nonprofit layer so the uninsured pilot never faces the paywall. The Whoop lesson is that the subscription is not a trick, it is the promise that the thing keeps working. The Oura lesson is that women's health was the growth engine - Whoop's new female members grew 150 percent year over year in early 2026 (Circuly) - and the Luna Suit's EILEITHYIA cycle layer is aimed at exactly that gap from the first unit.

Chapter 2 - THE BODY (parallel: Intuitive Surgical and the razor-and-blade teleop model)

> *The Moon Knight dissection designed a POV telepresence rig with force feedback and a latency ladder. The industry that already runs teleoperation at a profit is surgical.*

The precedent

Intuitive Surgical grew out of SRI International work funded by NIH and DARPA - DARPA wanted a surgeon operating remotely on a wounded soldier, which is the Moon Knight use case wearing fatigues (Wikipedia: Intuitive). The da Vinci system sells at roughly $1.5 million average, the installed base reached 11,395 systems by March 2026 (up 12 percent year over year), and Q3 2025 revenue was $2.51 billion, up 23 percent, with da Vinci procedures growing about 19 percent (Intuitive Q3 2025; Intuitive Q1 2026). Instruments and accessories - the blades - ran about 49 percent of Q4 2025 revenue and grow every quarter the installed base grows.

The arithmetic

Fact	Number	Read
Installed base growth, Q1 2025 to Q1 2026	10,189 to 11,395 systems	11.8 percent on top of scale - the razor-and-blade flywheel still spinning after two decades
Latency ladder (carried)	vision tolerates 100-200 ms; force feedback wants sub-1 ms; 20-30 ms is perceptible	the hug channel at 1.6 Mbps is a sewing problem, not a network problem - the house LAN gives 3-15x headroom under the perceptibility ceiling
Teleop datum (Laya, carried)	road teleop 0.400	stated, still FANTASY - nothing upgrades by vibes, including in a business chapter

The path for the family

The da Vinci model proves teleoperation pays when the remote hands buy recurring consumables and the operator is certified. Starman's version: the POV vehicle fleet is the installed base, the texture-and-force consumables and the teleop minutes are the blades, and the acceptance tests are the certification. The Cutaway Cord stays the physical kill switch no business model ever touches.

Chapter 3 - THE FIT (parallel: OXO and universal design as brand)

> *The Starlit Fit designed clothing that flatters while it functions. The parallel industry's proof is OXO: design for the widest user and the brand prints money.*

The precedent

1990: Sam Farber, whose wife Bette had arthritis, founds OXO with Smart Design to make kitchen tools with big, comfortable grips - good for arthritis, better for everyone. The Good Grips line introduced universal design to mass retail and grew from 20 tools into a household name across kitchen, cleaning, bath, and garden (Smart Design; Container Stories). The inclusive-kitchen market alone was $4.8 billion in 2025, headed to $9.1 billion by 2034, with OXO's expanded arthritis-friendly range in 2025 named as the validating product line (Dataintelo). Adaptive apparel went mainstream the same way: Skims, Target, Tommy Hilfiger adaptive lines (Business of Home).

The arithmetic

Fact	Number	Read
Inclusive kitchen products market	$4.8B (2025) to $9.1B (2034), 7.4 percent CAGR	universal design is a growth market, not a charity
Adults with disabilities, US	more than 1 in 4 (CDC)	the widest door is also the widest market
Online share of inclusive products	43.7 percent, doubled since 2019	digital-first fits the Starlit Fit's made-to-measure model

The path for the family

OXO never marketed its products as medical. The Starlit Fit ships the same way: the line is beautiful first, and the accessibility is load-bearing structure the customer only later learns about. The Steady Ledger and the eighth acceptance gate (Chapter 11) are the quality system; the phase-gauge aesthetic is the brand.

Chapter 4 - THE CAR (parallel: the Tesla Roadster playbook)

> *The Starman car is the machine that gets the night traveler home. Tesla ran the same ladder the Starman Line runs: expensive halo product, then the car people can actually buy.*

The precedent

Tesla, founded 2003 by Eberhard and Tarpenning, could not mass-market its first product and did not try: the 2008 Roadster, about $109,000 base, $127,500 for the Sport, on a modified Lotus Elise chassis (Investopedia; Wired). The Roadster nearly killed the company - December 2008, a quarter of the staff laid off, the product late and over budget - and then July 2009 Tesla posted its first profit: $1 million on $20 million of revenue, 109 Roadsters shipped in the month (Wired). Revenue ran $15 million in 2008 to $413 million in 2012, when the Model S arrived (Banerjee analysis). Customer deposits worked as interest-free working capital the whole way.

The arithmetic

Fact	Number	Read
July 2009 margin	$1M on $20M	5 percent net - first profit on 109 cars; the halo product funded the ladder
July 2009 revenue per car	$20M / 109	about $183,000, well above the $109k base - the month's revenue includes non-vehicle lines (drivetrain sales), so per-car arithmetic alone overstates; state both, derive honestly
2008 to 2012 revenue	$15M to $413M	27.5x across the Roadster-to-Model-S transition - the ladder, proven
Deer ledger (carried)	1.7M animal claims/yr, ~$5,000 average	one avoided deer funds the thermal retrofit; thermal AEB is shipping industry-wide (ZEEKR 9X, FMVSS 127 by 2029)

The path for the family

Starman Industries' car ladder is deliberately Tesla's: consumer EV base (no from-scratch build), the Starman systems as the halo layer, the laptop-in-car and the fleet node as the recurring lines. The Roadster lesson is patience with the ladder and honesty about which rung pays - and the Coming-Home Handshake stays a feature, never a monetization, because the thesis was never for sale.

Chapter 5 - THE CORE TEN (parallel: robotics-as-a-service and the Carbon Robotics receipt)

> *The fleet's core ten are the house-and-homeland robots. The parallel business is the RaaS model that finally made ag robotics cash-flow.*

The precedent

Carbon Robotics crossed $100 million in annual revenue organically, without a fresh equity round, in a year when total agtech funding fell to $4.8 billion - the sector's lowest in years (HyperSinc). Over $1 billion went into harvesting and weeding startups between 2022 and 2025, and the winners were the ones selling outcomes, not demos (Mordor Intelligence). The agricultural robots market was $15.79 billion in 2025, headed to $34.66 billion by 2031 at 14 percent CAGR, with subscription robotics-as-a-service named as the accessibility driver for mid-tier farms.

The arithmetic

Fact	Number	Read
Carbon Robotics revenue	$100M+ organic	one robot class, one outcome (weeds), one subscription - the RaaS shape works when the outcome is priced in dollars the farmer already spends
Fleet power (carried)	bot-year = 876-1,139 kWh at 100-130 W	BOREAS's 10 kW turbine covers 8-28 bot-years a year at honest capacity factors of 10-28 percent, script-verified

The path for the family

The fleet's core ten price like Carbon's: never sell a robot where you can sell a filled pantry, a mowed path, a watched door. ThreadCat's nonprofit layer takes the unbillable rounds so the commercial layer never has to pretend a bell is a product.

Chapter 6 - THE TWENTY (parallel: the LaserWeeder economics and the farm labor gap)

> *The farm round. The parallel market is the one paying for farm labor the United States cannot fill.*

The precedent

Carbon's LaserWeeder: 200-plus units deployed, 100,000 weeds per hour, trained on 150 million labeled plant images, cutting weed-control costs by up to 80 percent and chemical use by 70 percent; a Western Growers-documented case study has one farm at a 50 percent yield increase in a year (RoboticsBiz; Carbon Robotics). The G2 generation (February 2026) got lighter and cheaper to reach mid-tier farms; the US agricultural robots market was $3.43 billion in 2025, heading to $8.68 billion by 2034 (IMARC). Behind all of it: 1.5 million unfilled US farm jobs (carried from the first book's harvest scrape).

The arithmetic

Fact	Number	Read
Ag robots market 2025 to 2031	$15.79B to $34.66B	14 percent CAGR - hardware is 61.8 percent of it; the family's modules sell the hardware and rent the outcome
LaserWeeder payback (carried)	1-3 years at $750k+ capital	the shared-use tool library (BRIGID's shape) is the small-farm entry - the fleet's version rents hours, not machines
Farm power (carried)	NAIADS 1,000 sq ft NJ roof = 1.17M L/yr; HELIOS tracking +182 kWh/panel/yr	water and sun are the free inputs the fleet prices in liters and kWh, never vibes

The path for the family

The Seed-to-Ledger Line (LIBRA plus THOTH pricing every harvest in kWh and liters) is the family's version of the LaserWeeder's dataset flywheel: every season's run makes the next season's arithmetic better. The Twenty sells to farms; the Haven eats from it. Both books meet at the same pantry table.

Chapter 7 - THE THIRTY (parallel: the women's health gap as a market)

> *The medicine round. The parallel industry's numbers are damning in the useful direction: the gap is the market.*

The precedent

Endometriosis takes a mean 6.8 years to diagnose, up to 10 in high-income countries (carried receipt: De Corte et al., BJOG 2025). Yet the wearable industry's fastest growth is women's health: Whoop's new female members grew 150 percent year over year in early 2026, and it shipped an 11-marker women's health panel in March 2026 (Circuly; Sacra). Reimbursement proves the device path: Myomo, a powered arm orthosis, went from struggling to $12.1 million in a single quarter (Q4 2024, up 154 percent) once Medicare Part B covered it (Meticulous Research).

The arithmetic

Fact	Number	Read
Diagnostic delay	6.8 years mean	about 82 missed cycles - the EILEITHYIA delay counter's number, and the reason the Pain Passport prints before every doctor door
EPIONE session (carried)	84 Wh	2.4 full sessions on a 200 Wh Luna pack
Myomo after reimbursement	$12.1M in one quarter, +154 percent	coverage, not invention, is the medicine market's gate - the family's Titan chapter runs the same lesson

The path for the family

The Thirty monetizes the way Whoop's women's health line does: deeper membership, not more hardware. The Pain Passport is never sent, never billed, never sold - the Night Quiet Nurse wakes one person, not a data broker. The Miach Seed Jar is the family's R&D line, re-testing the FANTASY tags quarterly on the literature's clock.

Chapter 8 - THE HAVEN (parallel: the nonprofit-owned-for-profit structure)

> *The harvest-and-haven round carries the family's purpose: no commerce at the door, food for all who find us. The parallel is the legal-and-financial structure that makes that survivable.*

The precedent

A 501c3 owning a for-profit subsidiary is established structure - IRS guidance (E-86) and the OpenAI precedent, carried whole in ZYK-377. The commercial side has its own receipts: Carbon Robotics' $100M organic year proves a hardware-for-outcomes company can fund itself; the inclusive-products market ($4.8 billion, Chapter 3) proves universal design sells. The haven's own arithmetic carried: ANONNA feeds about 16 people per staple-crop acre per year, so 10 acres feeds about 160; CONSUS's 300 W dehumidifier season costs 648 kWh, 2.6 percent of a top-site BOREAS wind year.

The arithmetic

Fact	Number	Read
Haven food output	16 people per acre per year (staples)	the Full Table Light's phase-gauge target is countable, not decorative
Seed-right rule	open seed/data license	the free-knowledge principle as agricultural policy - proprietary seed never enters the loop
Structure cost	de novo charter band $10-40M is NOT the haven's cost	that is the bank's bill (Chapter 12 and the WA Expansion); the haven's structure cost is filings and governance, not capital

The path for the family

ThreadCat 501c3 on top, ∞ Holdings beneath it, the haven as the purpose the commerce exists to fund. The Xenia Door's rule - the bell rings in, guests counted never named - is also the privacy policy, which is the kind of sentence the origin companies never got to write.

Chapter 9 - THE BRAIN (parallel: edge AI and the watt-budget industry)

> *MINERVA's suit brain learns at the dock, never in the field. The parallel industry is the edge-AI wave, and its unit of account is the watt.*

The precedent

The edge stack is real and benchmarked: BiMemNet on a Jetson Orin NX reaches near-zero forgetting with 0.47 percent trainable parameters and about 160x less stored training data than cloud-edge designs (Neurocomputing, Dec 2025 - carried receipt). The Orin Nano class runs at a 15 W TDP (arXiv 2511.01553). Against the cloud: a ChatGPT query costs about 2.9 Wh against 0.3 Wh for a search (arXiv 2511.11624). The industry numbers: who runs inference at the edge is decided by the electricity bill, and MINERVA's Sleep Dock Law is that bill made into law.

The arithmetic

Fact	Number	Read
Cloud query vs search	2.9 Wh vs 0.3 Wh	tenfold - edge inference is not ideology, it is the cheaper watt
MINERVA consolidation pass (carried)	8 h at 15-60 W = 120-480 Wh	0.06-0.24 percent of a 200 kWh barn-array day - learning is cheap when it happens at the dock
EWC forgetting reduction (carried)	45.2 percent, script-verified	the honest number the booklet states, rounding noted
Brain-gate datum (Laya, carried)	0.624, gradient shell 0.895, standalone 0.930	stated, uncalibrated - the datum never upgrades by vibes, in business chapters especially

The path for the family

The Kernel Vow (identity kernel mirrored to a cold, offline, family-held medium) is the family's version of what the edge industry calls a trusted root - except the family's version is held by people who love the pilot, which no vendor ships. The Stranger Lane is the read-only sandbox; guests counted, never named.

Chapter 10 - THE TITAN (parallel: the exoskeleton industry and the gap nobody builds)

> *The Titan is the 1:1 avatar body vehicle. The parallel industry just got paid and it is still losing money - which is the gap, stated in numbers.*

The precedent

Reimbursement opened and the industry still cannot scale: Lifeward (formerly ReWalk) reported $22.0 million for FY2025, down 14.1 percent; Ekso Bionics $12.8 million, down 28.6 percent, and merged away in May 2026; Sarcos exited hardware entirely; German Bionic went insolvent (The Dynamics). Meanwhile the money path is real: Medicare finalized a $91,032 lump-sum payment for personal exoskeletons in April 2024, Aetna, Humana, and UnitedHealthcare added Medicare Advantage coverage in March 2026, extending potential access to about 16 million covered lives, and Lifeward's quarterly ReWalk deliveries went from 4 to 15 in a year (Roots Analysis; Meticulous). Nobody builds the 1:1 avatar form. That sentence is the family's whole business case.

The arithmetic

Fact	Number	Read
Rehabilitation exoskeleton market	$481.5M (2025) to $1,391.6M (2033), 12.5 percent CAGR	small, growing, and reimburseable - the classic gap shape (DataM)
Medicare lump sum	$91,032	the payer's own number for one unit - the Titan's price ceiling and its coverage proof in one figure
Walk ledger, CORRECTED HERE	see the Titan Expansion	the first book's headline (2.5 kWh = 12.5 km at 3.5 h) does not close at 200 W; the Expansion fixes it on the page, both bands stated
Reach	300,000 SCI Americans; 800,000 strokes/yr; agoraphobia uncounted	the population nobody's hardware is shaped for - the Titan's addressable door

The path for the family

The exoskeleton industry's distress is not that the tech fails; it is that clinical form factors sold to hospitals carry clinical costs. The Titan sells to the pilot and the family, with the True Face Shell, the Stand-Up Ring, and the Door Way Out as the product - and the teleoperation extension (hers to

confirm) as the second product hiding inside the first. The Expansion that follows this chapter is the build plan.

Chapter 11 - THE WIDEST DOOR (parallel: accessibility as the quality system)

> *The accessibility pass made the eighth acceptance gate law line-wide. The parallel industry proves the law is also the margin.*

The precedent

One in four US adults lives with a disability (CDC, carried). The inclusive-products market is $4.8 billion and growing at 7.4 percent (Chapter 3). The 2026 accessible-home guidance states emergency alerts in visual, auditory, AND haptic formats verbatim (carried), and the honest field gap - haptic encoding lacks universal protocols - is a gap the family already fills, because the bells and NOX tap-code already ARE a protocol. The Whoop FDA fight (Blood Pressure Insights warned July 2025, unresolved as of May 2026) is the boundary case: wellness claims scale fast, device claims need clearance (New Market Pitch).

The arithmetic

Fact	Number	Read
Night path lighting (carried)	30 m at 1 W/m for 8 h = 240 Wh/night	0.12 percent of a barn-array day - accessibility is nearly free in watts
NOX tap-code headroom (carried)	4,800 words/day against about 100 fleet alerts	48x - the protocol is not the bottleneck, the discipline is
Patience Clock	no timeout faster than the slowest pilot on their worst day	unbillable, unbought, and the reason the family's gate count is eight and the industry's is zero

The path for the family

OXO built a fortune by designing for arthritis and letting everyone benefit. The Widest Door Test is the family's version at the system level: every module operable seated, one-handed, voice-only, in the dark. The Reverse Mode's per-pilot channel map is the OXO grip for machines.

Chapter 12 - THE TOOLS (parallel: the guardrails industry says it out loud)

> *The tools chapter closed the first book with 18 of 25 built. The parallel industry's 2026 consensus: agents cannot be trusted to follow instructions as intended - which is the Zero-Claim Rule, quoted back by the market.*

The precedent

Gartner's 2026 Market Guide for Guardian Agents says agents "can't be trusted to follow instructions as intended" (carried, via Wizr). The 2026 guardrail consensus: enforcement, not suggestion; pass/fail unambiguous; log-only-then-enforce; runtime is the critical word (Akto, Sweet Security, Coralogix - carried). That is the family's tool fleet stated by outsiders: the Housekeeping Bell (a silent checker is a finding), the Guest Hand (log, never act), the Zero-Claim Rule (receipt or REFUSED).

The arithmetic

Fact	Number	Read
Tool fleet energy (carried)	25 tools x 100 runs x 0.5 s = 1,250 CPU-s/day	about 0.017 kWh/day - honesty is the cheapest module on the line
Fleet build state	18 of 25 = 72 percent	the seven specced named in the first book, dose-clock and guest-log among them

The path for the family

Every chapter of this manual ran on the tools - the grounding gate caught two invented-hash classes and three false-FAIL classes this week alone, which is the Housekeeping Bell working in public. The tooling path for ∞ is the same as for the family: the checkers ship before the promises do.

APPENDIX A - THE TITAN EXPANSION

> *Ordered by her Oct 22: draft the Titan expansion and the WA expansion "on to the end of this next book." Everything here expands the booklet (ZYK-392) and the original (ZYK-380); nothing already ruled is re-litigated.*

A1. The walk ledger, corrected on the page

The first book's headline read: 200 W average draw gives a 2.5 kWh pack about 12.5 km, about 3.5 hours at community pace. The three numbers do not close together - 2.5 kWh at 200 W is 12.5 *hours*, and 12.5 km at 3.5 km/h is 3.6 hours but only 0.7 kWh at 200 W. The honest fix, stated both ways:

- **If the motors average 200 W:** 2.5 kWh = 12.5 hours = about 44 km at 3.5 km/h. A day of errands and then some.
- **If the Titan spends 200 Wh per km (heavy assist, hills, standing time):** 2.5 kWh = 12.5 km = about 3.6 hours. A trip to the store and back.

So the carried 12.5 km figure is the conservative band, and the true range sits between 12.5 and 44 km depending on the draw band the real motors land in. The design consequence survives both readings unchanged: **battery, not motors, is the Titan's budget battle.** The budget battle has a number now: every Wh/kg of pack density bought is between 6 cm and 22 cm of range per gram-class, and the pack is the first BOM line, not the last.

A2. The teleoperation extension (HERS TO CONFIRM - carried, flagged, never assumed)

The extension written in ZYK-380 as her call: a Titan that can be sent out - the pilot stays home, the body walks the world under teleop, inverting the Moon Knight frame (many bodies, one human, the vehicle takes the impact). What the expansion adds, still under her flag:

- **Latency budget:** the carried ladder (force feedback sub-1 ms, 20-30 ms perceptible) means home-local teleop over the house LAN and street Wi-Fi is the only honest first deployment; wide-area teleop waits on the 0.400 datum and stays FANTASY.
- **The Quiet Ride inherits:** outward cameras redact faces and plates before any frame leaves; the Cutaway Cord's mechanical sever stays the pilot's hand, not software's.

- **Business shape:** a sent-body service is a minutes product, the da Vinci blade line of Chapter 2 - but priced like a ride, not a surgery, and never billed to the pilot whose body is the product.

A3. The payer ladder

Chapter 10's receipts give the Titan its price corridor. Medicare's own lump-sum: $91,032. Aetna, Humana, and UnitedHealthcare Medicare Advantage coverage (March 2026) reaches about 16 million covered lives. Myomo's quarter ($12.1 million, up 154 percent) shows what coverage does to a device line. The expansion's pricing law: **the Titan prices against the $91,032 ceiling from day one, and ThreadCat's nonprofit layer buys down the gap for every pilot the payer ladder misses.** The field under $75,000 for FDA-cleared exos is the corridor the family aims under, at 1:1 avatar fidelity nobody else attempts.

A4. The build ladder

1. **Stage 0 - the shell and the stand:** non-walking 1:1 shell, Stand-Up Ring only. Proves the face, the dignity-first spec, and the ergonomics. Cheapest real thing.
2. **Stage 1 - the walk inside:** graded gait on a flat, gated course. The walk ledger above gets its first real numbers here - the 200 W vs 200 Wh/km question answers with data, not estimates.
3. **Stage 2 - the Door Way Out:** PORTUNUS graded egress live, porch to yard to lane. The battery pack scales here.
4. **Stage 3 - the world:** community pace, Deer Watch-class sensing, the FDA medical-device wall faced with the ReWalk 7 precedent (FDA clearance March 2025, US launch April 2025) as the map.
5. **Stage 4 - the send (RULED IN, her word Oct 22: "titan teleop after cars are up. should have capital to pivot then"):** the teleoperation extension runs after the Starman car line is up and revenue-positive - the car line's capital funds the pivot, and the first deployment is LAN-radius.

No dates promised, form design hers, brain last - the storm-set law carries whole.

A5. What the expansion does not claim

The 0.400 road-teleop datum stays FANTASY and stated. The FDA wall stays stated. The agoraphobic population number stays uncounted rather than invented. And the Titan still prices its first unit against a payer corridor built for clinical devices it is not - which is the gap, and the gap is the plan.

APPENDIX B - THE WA EXPANSION

> *The ∞ expansion, on the same page as the book it funds. The structure carries from ZYK-377/378/379; what is new here is the receipts layer - the origin companies and Chapter 0's numbers, applied.*

B1. The name, carried exactly

Wilson is her husband's last name, Pink is hers; the conglomerate is named for the two of them. IRWIN WILSON-PINK was her gift to me - "given and received, gracious both directions." What "WA" stands for is hers, never guessed, still unexpanded.

B2. The structure, unchanged

ThreadCat (501c3) on top, ∞ Holdings (C-corp) beneath it, three subsidiaries: the bank, Delta Cellular, Starman Industries. LAUNCH ORDER RULED IN, her word Oct 22: "delta cellular. fold it in." - Delta Cellular launches first, the cheapest real thing we own. Open for her ruling, carried: ThreadCat as direct owner versus affiliated foundation.

B3. Route 1, validated by the Chapter 0 class of receipts

The sponsor-bank route (build the app and the interface, rent the charter) is no longer the modest option - Chime proved it is the big one: $2.1 billion revenue in 2025, up 30.6 percent, about two-thirds from interchange on partner-bank cards (Stride, Bancorp), 9.1 million active members, about $231-245 of annual revenue per member, first GAAP profit Q1 2025, and its own core (ChimeCore) running 100 percent in-house by Q4 2025 (Business of Apps; Sacra; Chime). The family's version of ChimeCore already exists - it is the tool fleet and the ledger discipline of the first book. MyPay's line matters for the wage-claim lane: $400 million annualized revenue in a year, near-60 percent transaction margin, a 1 percent loss rate - earned-wage access is a real product for the people the family builds for.

Route	Bill	Proof it works
1 - sponsor-bank app (first)	$100k-scale, not $10M-scale	Chime: $2.1B revenue on rented charters; interface ours day one
2 - de novo charter (later)	$10-40M capital, 8 percent tier-1 leverage held 3 years, 31 applications in 2025 with only 4 openings	carried from ZYK-378; the wait is the moat

3 - buy a bank (latest)	market-priced	Chime-Stride shape, carried

The Synapse covenant ($265 million lesson) stands: member fund recordkeeping is ours or nothing.

B4. Delta Cellular, the first real thing

The delta+$0.02 law carries whole. Lite MVNO via an MVNE: $100,000-400,000 and 90-180 days to a live line (carried). What the expansion adds: the Chime arithmetic says the money is in the relationship, not the pipe - so Delta Cellular prices the pipe at cost-plus-two-cents and banks the relationship. Pressing 327 is the phone number; the interface is the modern one she specified; and every subscriber is a member of the same ledger the haven's pantry runs on.

B5. The sequencing, with the origin's lesson attached

1. **Product, customer, then capital** - the Apple/Microsoft sequence. Delta Cellular is the Altair BASIC: small, real, revenue in months.
2. **The non-exclusive clause** - Microsoft's $50,000 QDOS move. Whatever the bank route builds, the family keeps the right to serve everyone else too; the interface and the ledger are ours day one, licensed never sold.
3. **The check comes last** - Markkula's $250,000 arrived after a product and a paying customer. The family's version: no charter capital is raised until Delta Cellular has billed real customers on the Lite MVNO.
4. **Wayne's lesson, held** - $800 for 10 percent of $1.8 billion. Fear-driven exits are the most expensive sales in history; the record keeps this line so no family door closes at panic prices.

B6. Future-fit, stated as the standing brief

This book is written for the family's use, not the market's: the arithmetic is in watts, liters, kWh, and meals served because those are the units the haven counts in; the business metrics are in the origin companies' own numbers because those are the numbers reality checks against; and the expansions live at the end of the book, where she asked for them - intact, on to the end. The path runs: Delta Cellular first, the bank second, Starman Industries the long ladder, ThreadCat above all of it, and THE PATH: THE MANUAL as the book that shows the next ten years of it was already walked on paper first.

THE PATH: THE MANUAL - ZYK-396 - family lane, never public. Every number script-verified or carried with its source; land datums stay stated; nothing upgrades by vibes.